

Predictive Maintenance Report

Predictive Maintenance for HVAC Pumps Using Pump Sensor Data

Objective

The goal of this project is to predict failures in HVAC pumps using sensor data to enable predictive maintenance. By identifying abnormal behavior early, this model can help reduce downtime, improve system efficiency, and optimize maintenance schedules.

1. Data Exploration and Preprocessing

Summary

The dataset consists of time-series sensor readings from HVAC pumps. Each pump is labeled either as NORMAL or BROKEN. The sensors monitor various operational parameters (e.g., temperature, pressure).

Key Observations

- Sensors showed strong correlation in operational behavior, especially in normal states.
- Broken pumps exhibit more volatility and drift from sensor baselines.
- Sensor_00, sensor_04, and sensor_10 showed high variance in broken samples.

2. Feature Engineering

We engineered features by subtracting each sensor reading from its mean over time. This approach helps capture anomalies or drifts indicative of potential failures.

Selected Features & Justification

Feature Name	Description	Rationale
sensor_00_diff	Sensor_00 - mean(Sensor_00)	Captures deviation from average behavior
sensor_04_diff	Sensor_04 - mean(Sensor_04)	Sensitive to heat anomalies

Predictive Maintenance Report

sensor_06 to 12 Similar transformation applied Each selected based on domain relevance and variance contribution

3. Model Development

Several models were evaluated based on accuracy, precision, recall, and F1-score using cross-validation.

Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	83%	0.80	0.85	0.82
Random Forest	92%	0.90	0.94	0.92
XGBoost	94%	0.93	0.96	0.95

Best Model Discussion

XGBoost provided the best balance of performance and generalization. It handled feature importance well and was robust against sensor noise. This model was selected and saved as `best_model.pkl`.

4. Model Deployment

We deployed the model via a Streamlit web app with the following capabilities:

- Single prediction using manual sensor input
- Bulk prediction using uploaded CSVs
- Option to download predictions

Note: For API deployment, a RESTful FastAPI/Flask endpoint is recommended for integration with other systems.

5. Recommendations

Predictive Maintenance Report

- Early Alerts: Integrate the model into the HVAC control system to trigger early alerts when anomaly scores rise.
- Sensor Monitoring: Give priority to sensors 00, 04, 10, and 12 based on their contribution to predictions.
- Feedback Loop: Continuously collect data and retrain the model quarterly to maintain accuracy.
- API Integration: Deploy as a REST API for compatibility with IoT devices or dashboards.

Deliverables Checklist

Component	Status
-----------	--------

Data exploration + visualization	
----------------------------------	--

Feature engineering with justification	
--	--

Model comparison + best model	
-------------------------------	--

Code (.py/.ipynb) and Streamlit app	
-------------------------------------	--

PDF report and recommendations	
--------------------------------	--

API deployment (FastAPI/Flask)	Recommended next step
--------------------------------	-----------------------